



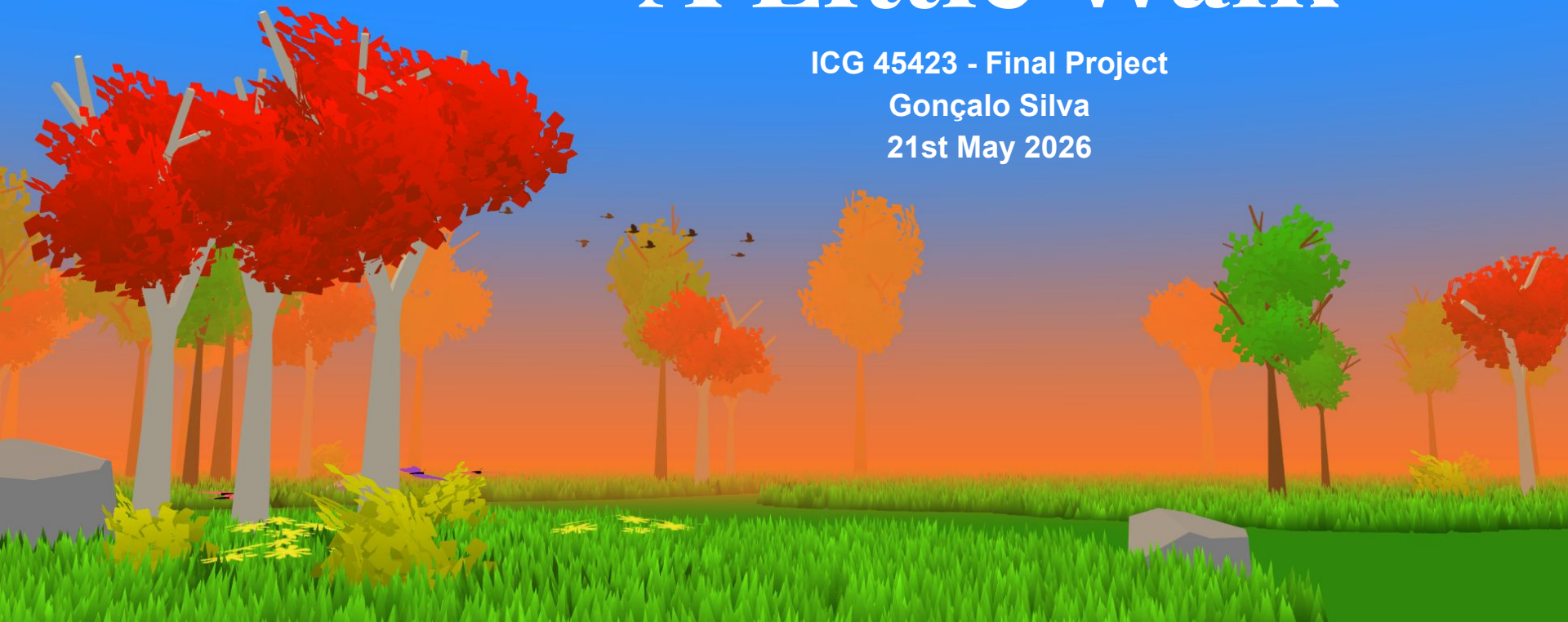
universidade
de aveiro

A Little Walk

ICG 45423 - Final Project

Gonçalo Silva

21st May 2026





a-little-walk.vercel.app

<https://github.com/goncalooliveirasilva/a-little-walk>

The Project

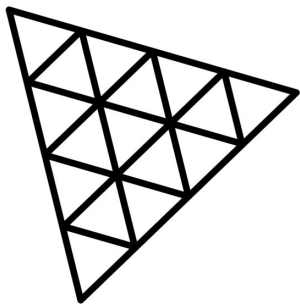
The Idea

Exploration game where the player controls a fox wandering freely through an open world.

- No concrete objectives
- Just exploration
- Low-poly, cartoon aesthetic

Inspiration: *The First Tree*, *ABZÛ*, *Alba: A Wildlife Adventure*

Technologies



(r183)



Tweakpane

stats.js

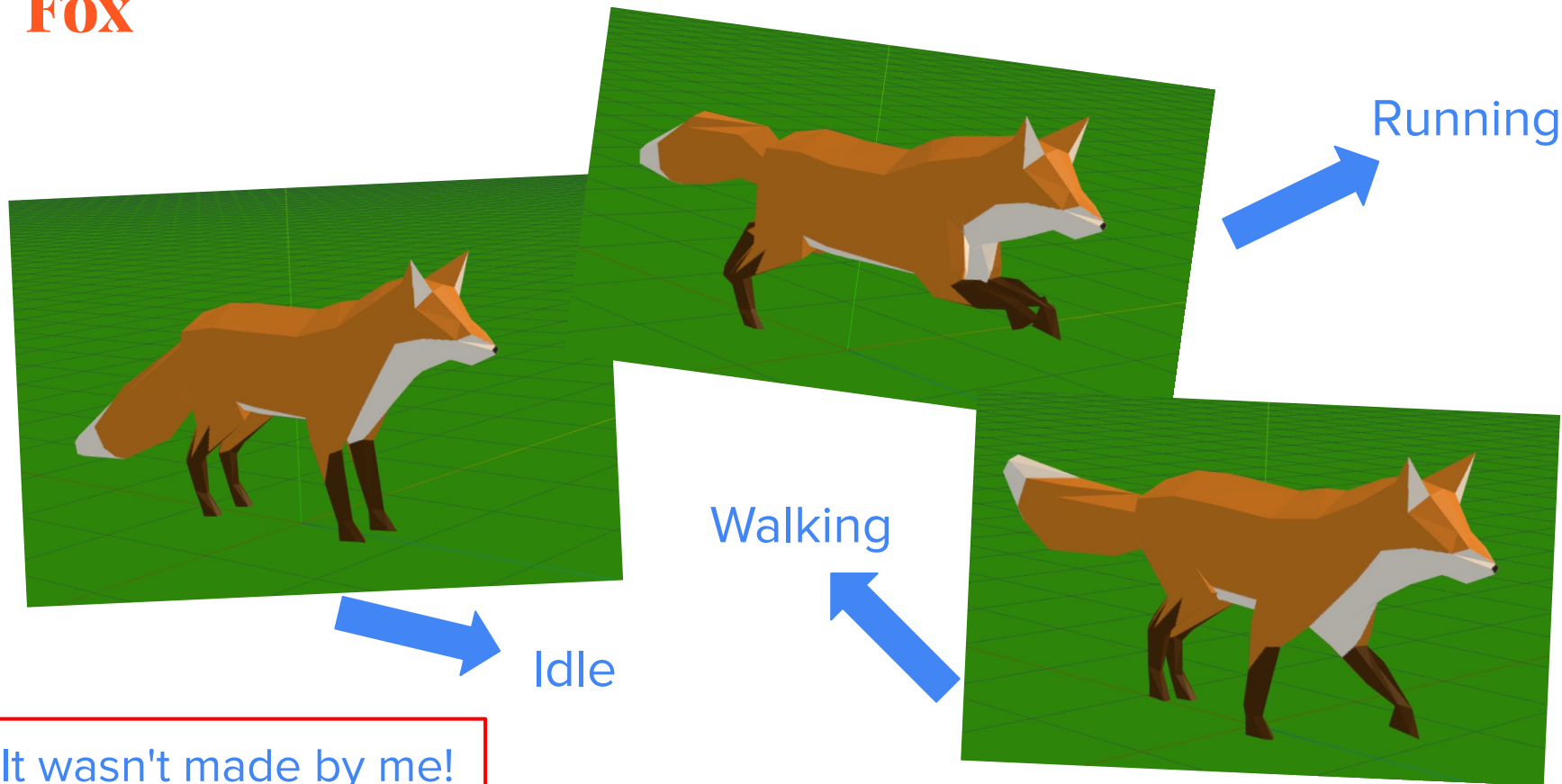
cannon-es



GSAP

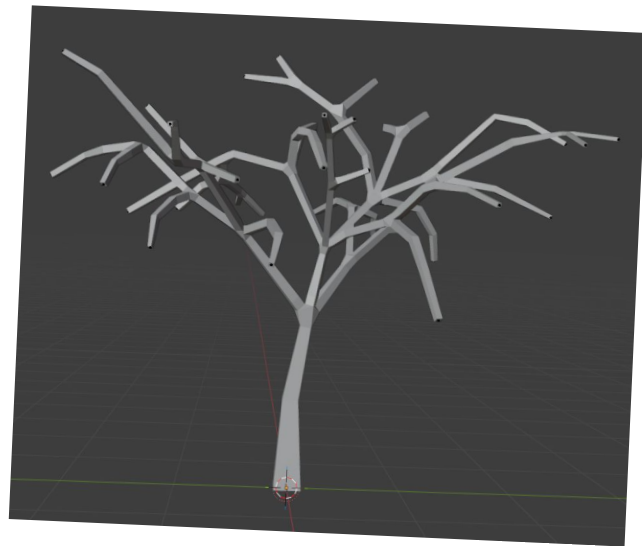
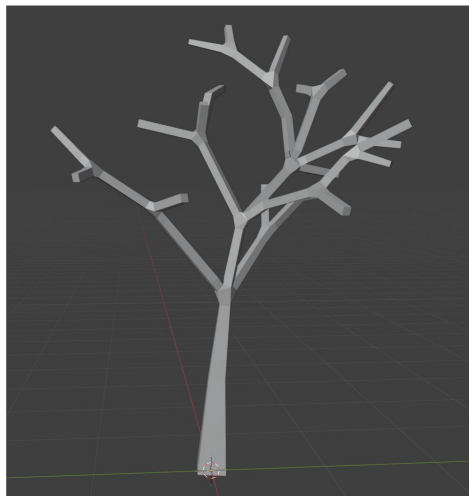
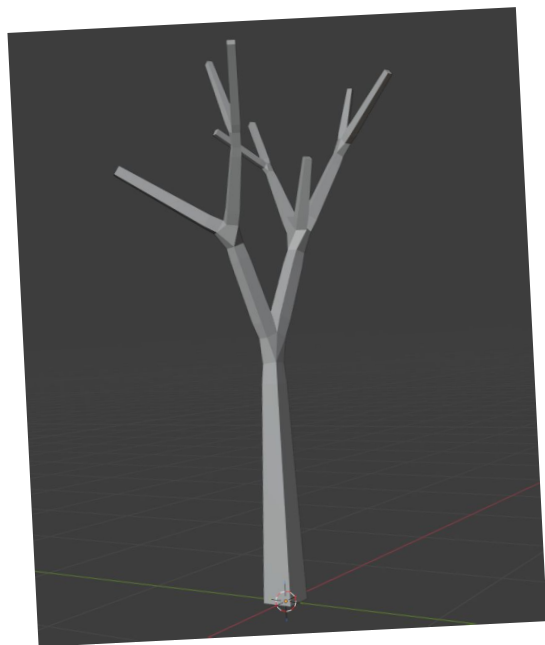
Models

Fox

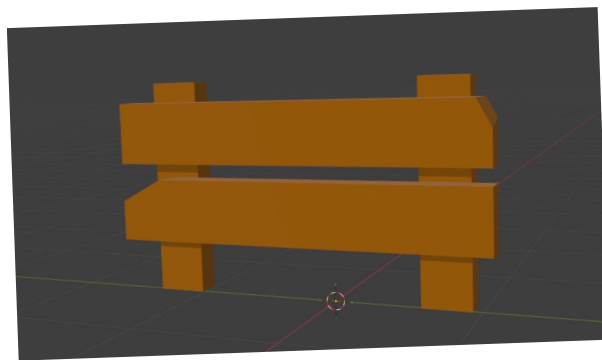
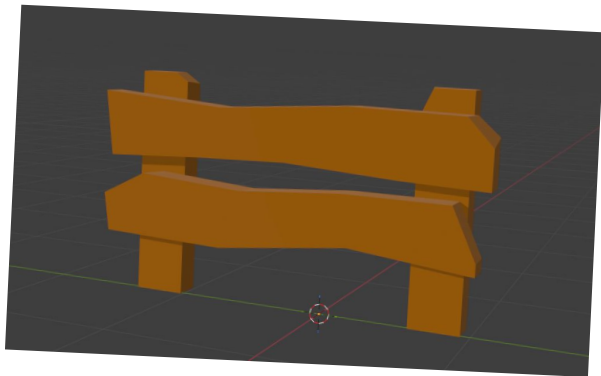
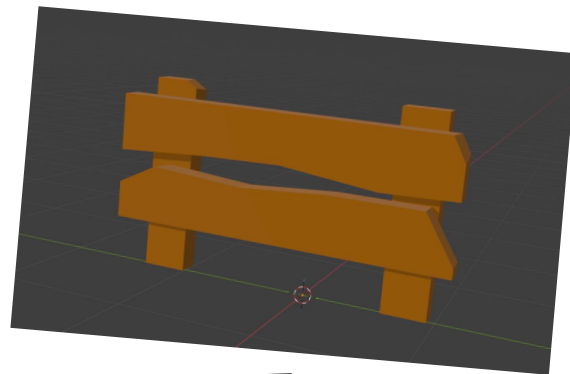
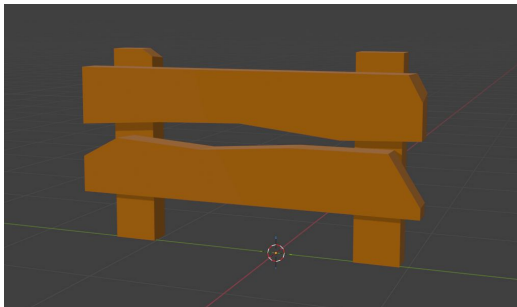
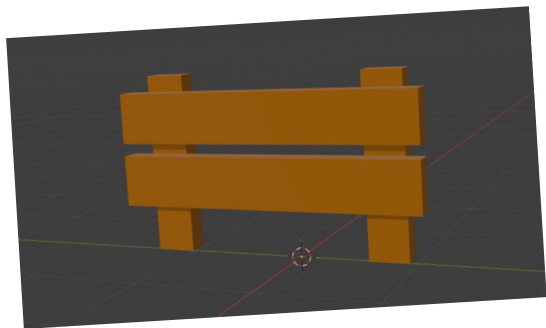


It wasn't made by me!

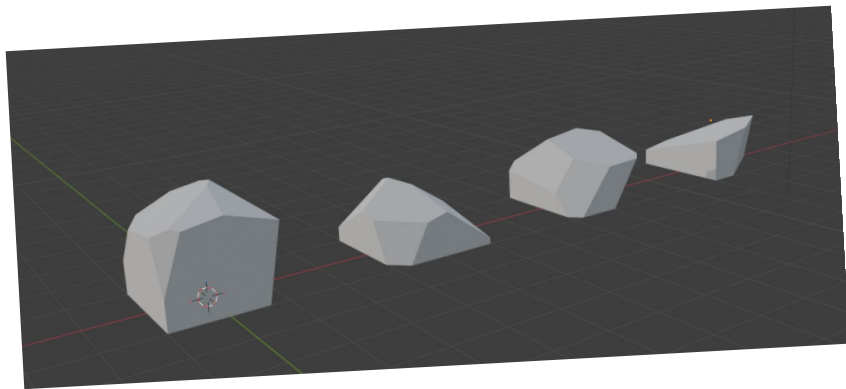
Trees



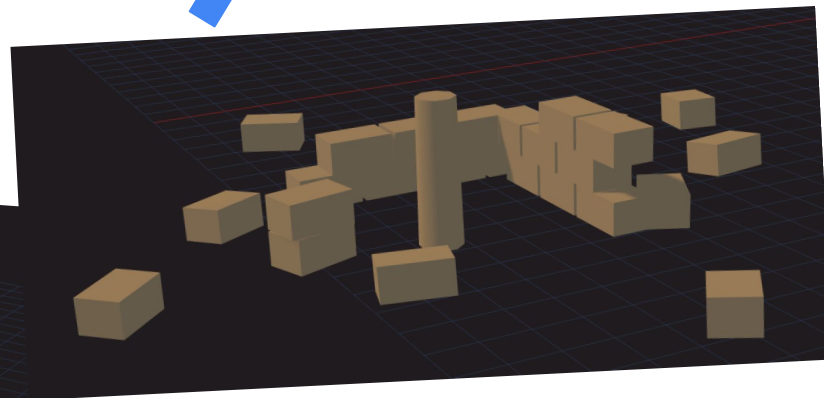
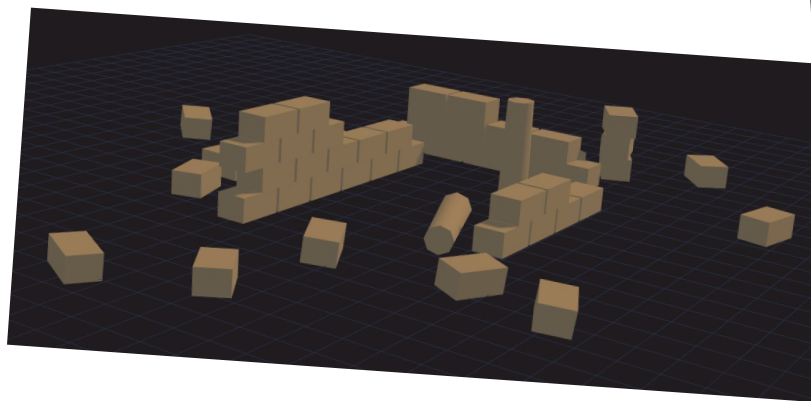
Fences



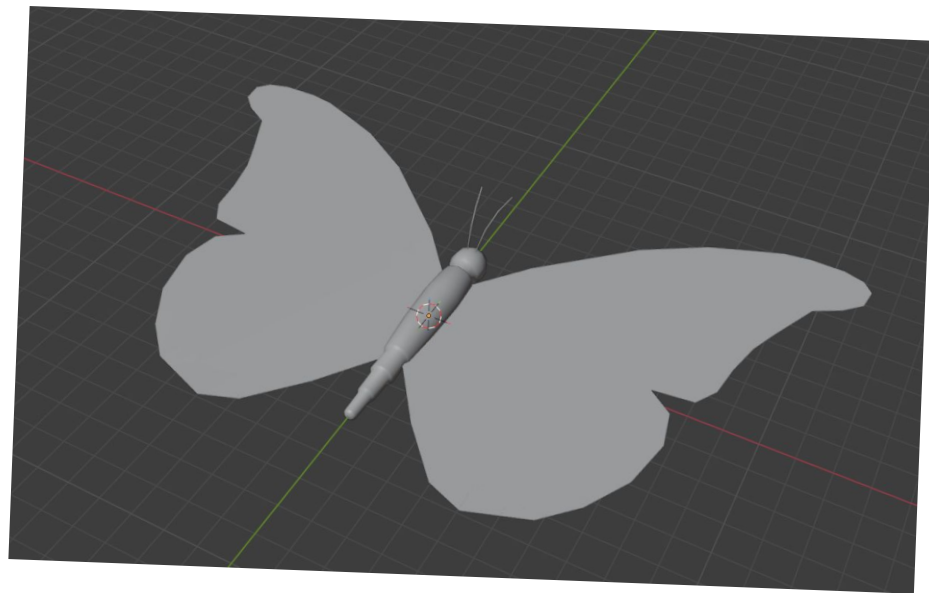
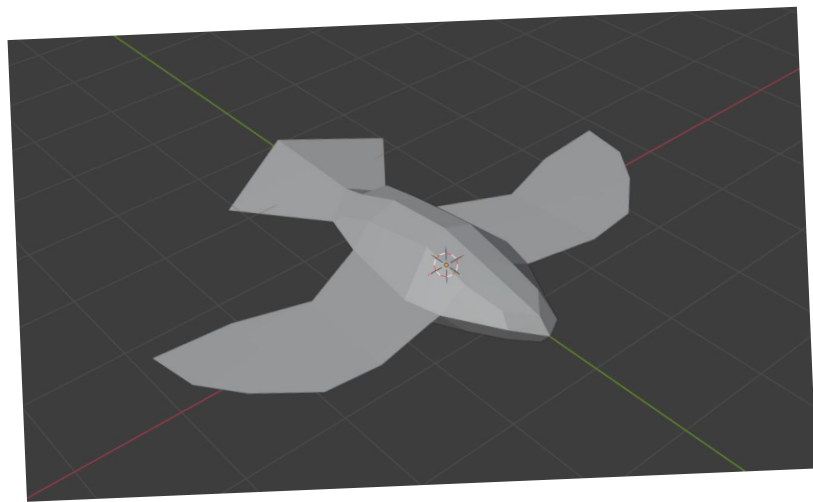
Rocks and Ruins



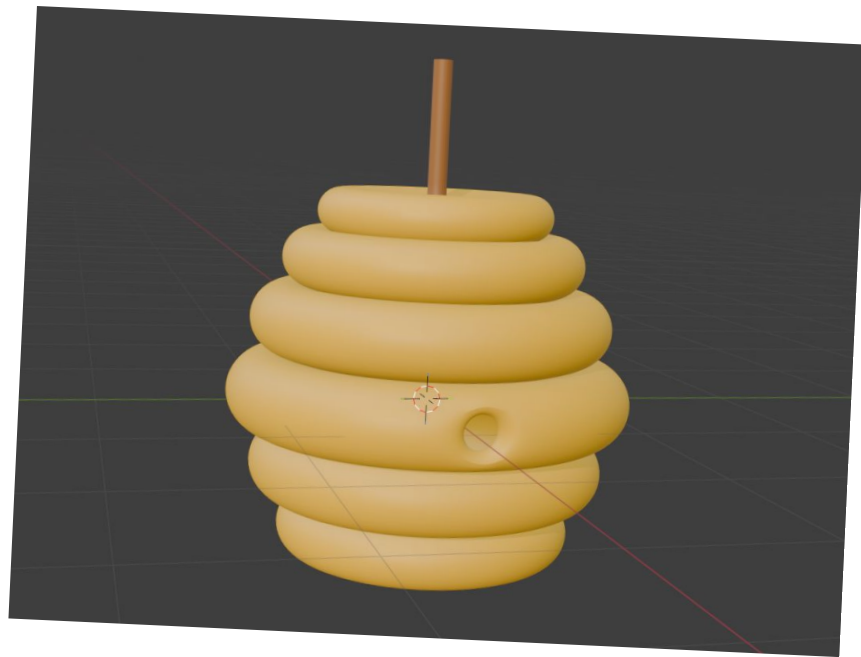
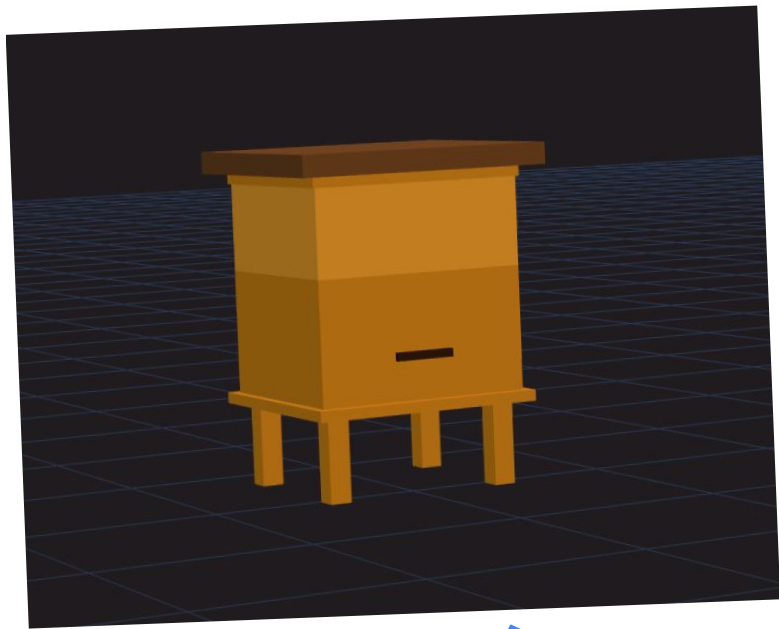
Made with primitives in
Three.js



Birds and Butterflies

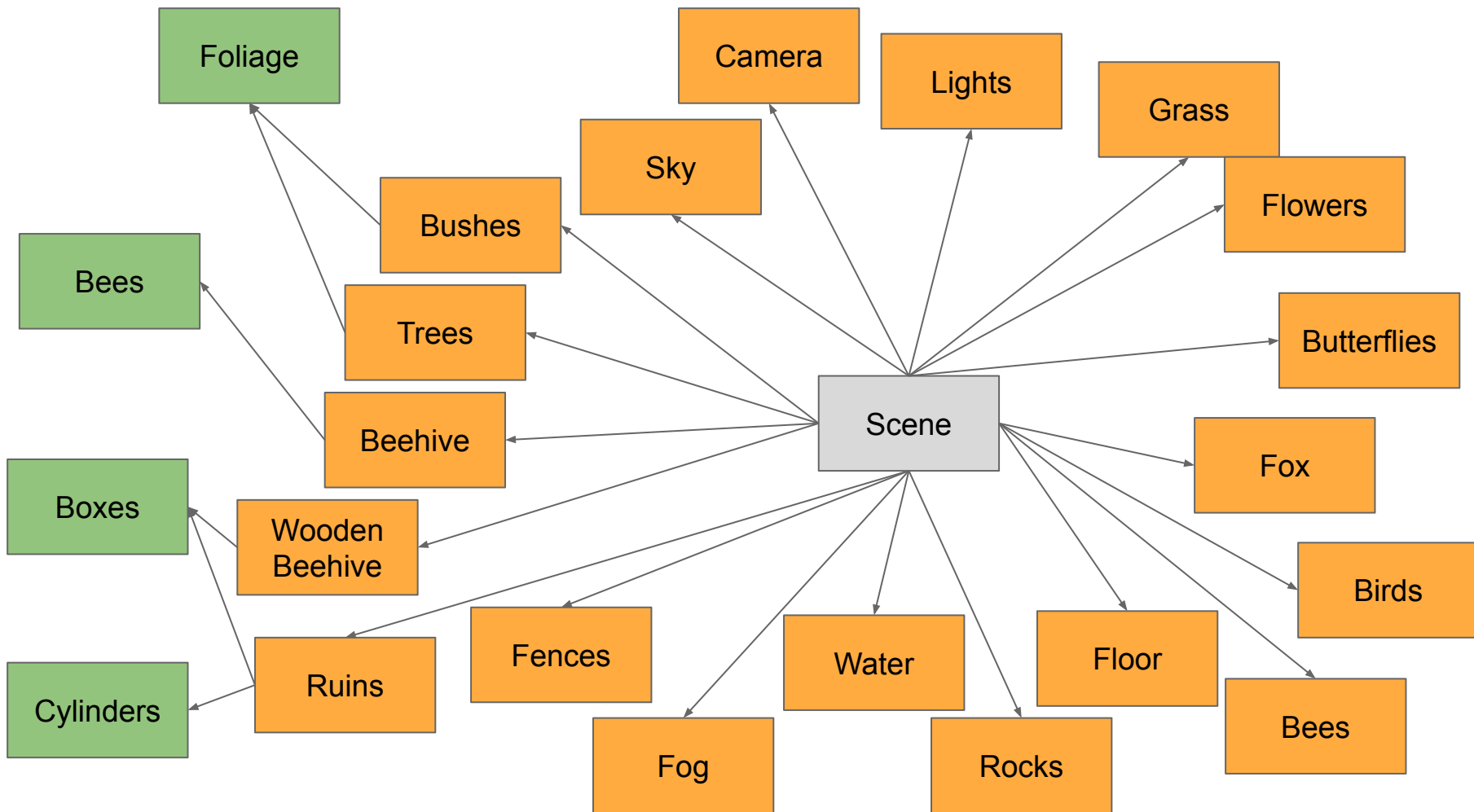


Beehives



Made with primitives in
Three.js

Scene Graph



Illumination, Materials, and Textures

Illumination

→ AmbientLight

→ DirectionalLight

```
1  setAmbientLight() {  
2      this.ambientLight = new THREE.AmbientLight("#ffffff", 1.4)  
3      this.scene.add(this.ambientLight)  
4  }
```

```
1  setSunLight() {  
2      this.sunLight = new THREE.DirectionalLight("#ffd08c", 2.5)  
3      this.sunLight.castShadow = true  
4      this.sunLight.position.set(30, 40, 20)  
5  }
```

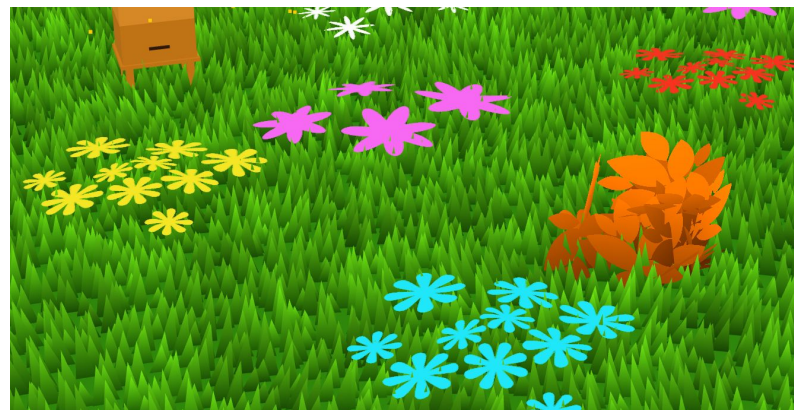
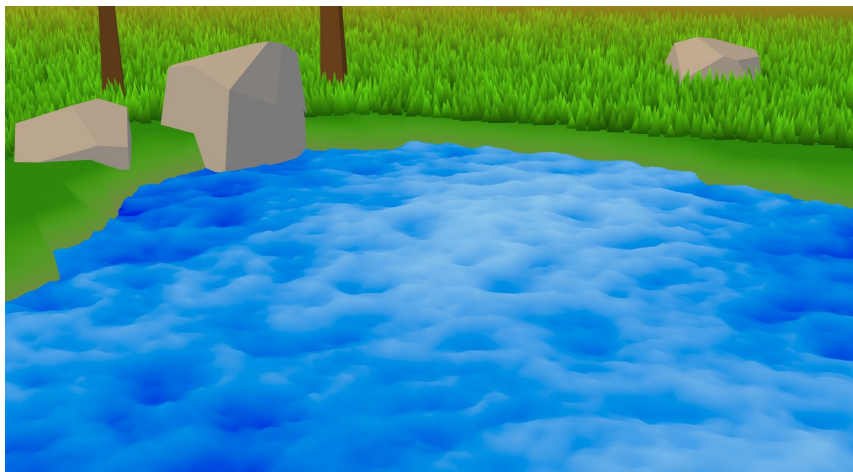
Materials

→ MeshStandardMaterial (wooden beehives and ruins)



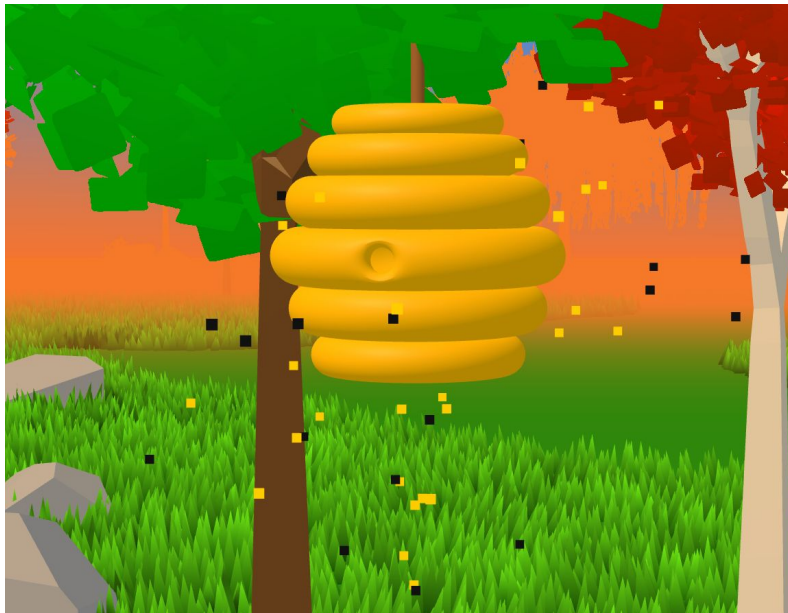
Materials

→ ShaderMaterial (grass, water, foliage, flowers, etc.)



Materials

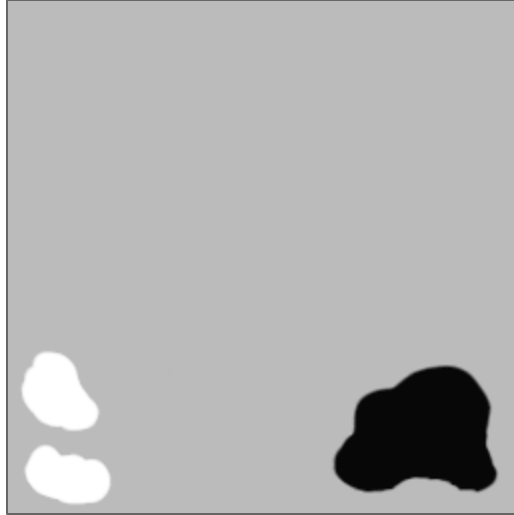
→ PointsMaterial (bees)



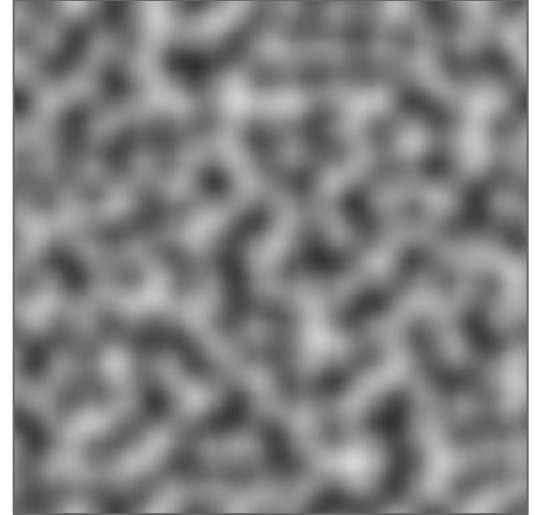
Textures



Texture to control the grass.



Texture to control the elevation of the terrain.

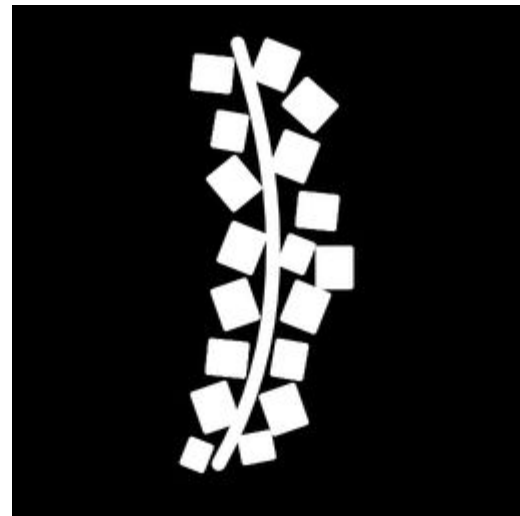


Perlin Noise Texture for wind effect.

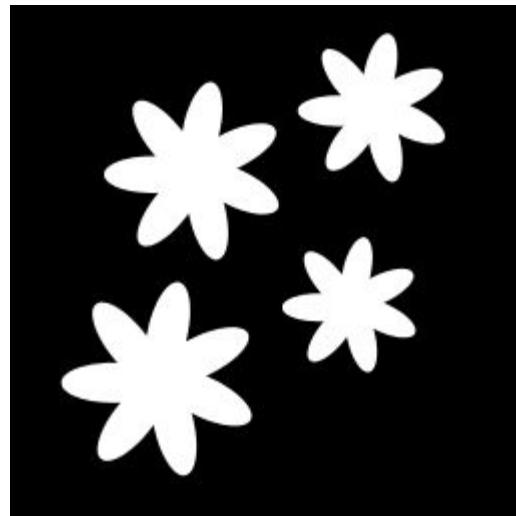
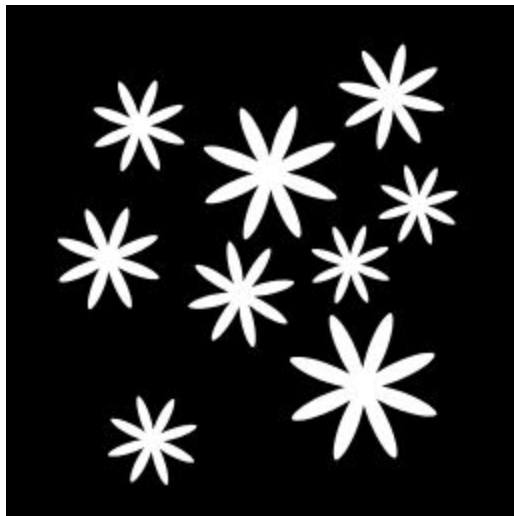
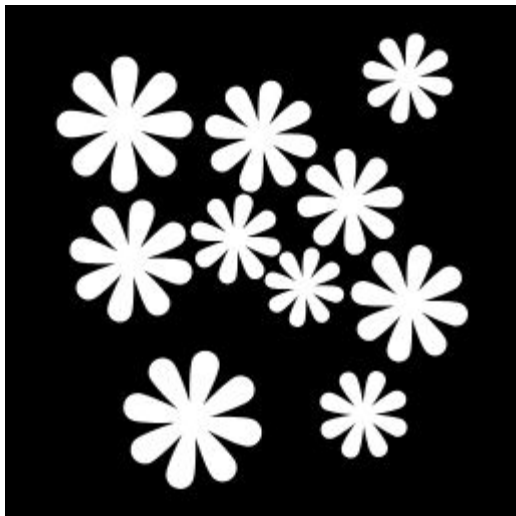
Textures made in Krita.

Perlin noise texture made in <https://gen3vra.github.io/perlinnoisegenerator>

Textures - Leafs



Textures - Flowers



Shaders

Flowers

Foliage

Water

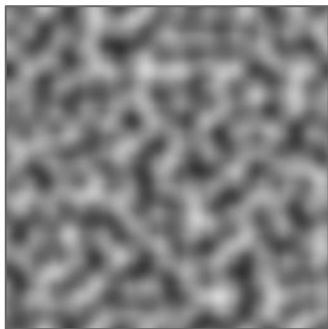
Sky

Willow

Floor

Grass

Shaders - Flowers



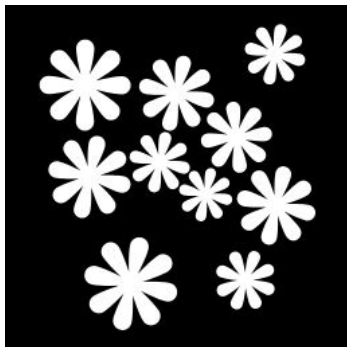
Perlin Noise Texture

Vertex shader

1. Red channel sampling with UV coordinates
2. Offset that value
3. Add it to the X and Z coordinates

Fragment shader

1. Passed UV coordinates of the vertex shader
2. Red channel sampling with UV coordinates
3. If the value is below a threshold, the pixel is discarded
4. Assign color



Flowers Texture

Animations

Animations

→ Fox (idle, walking, and running)



Animations already
included in the model

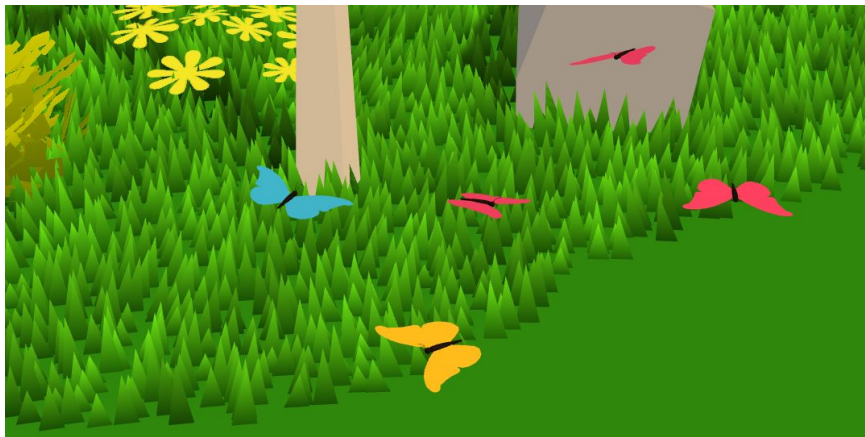


Animations

→ Birds

→ Bees

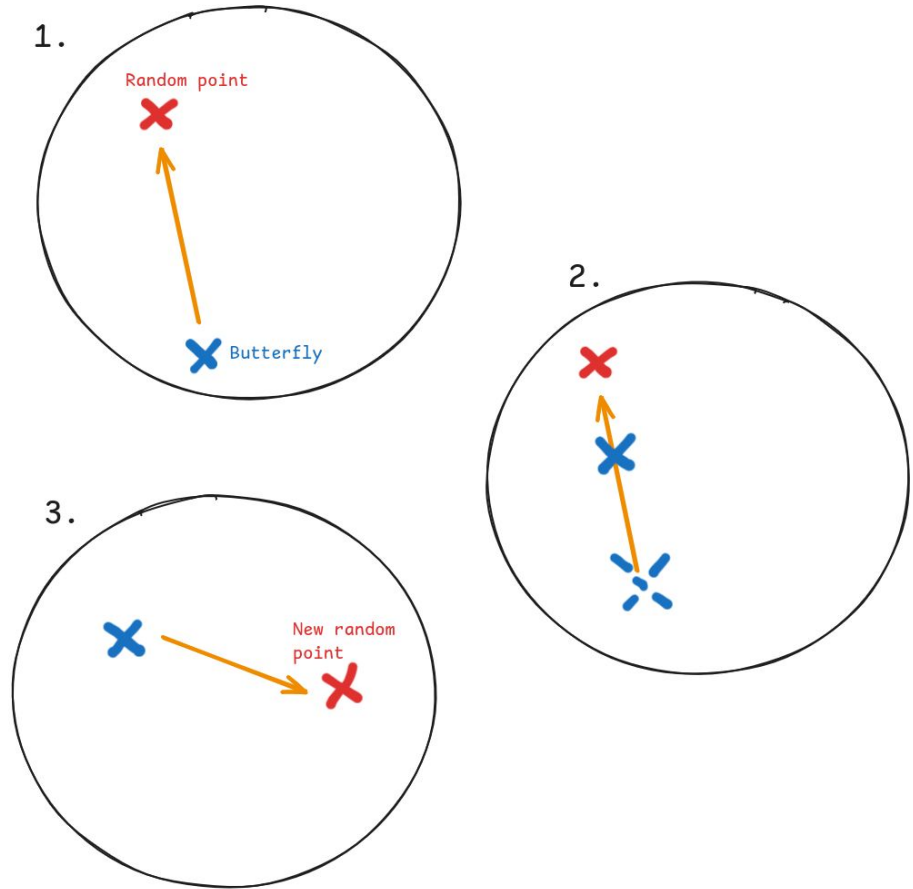
→ Butterflies



Animations

→ Butterflies

- ◆ Random point inside a radius
- ◆ Move toward it
- ◆ Pick a new one when closer
- ◆ Y bobbing with $\sin(\text{time})$
- ◆ GLTF animation for the wing flap

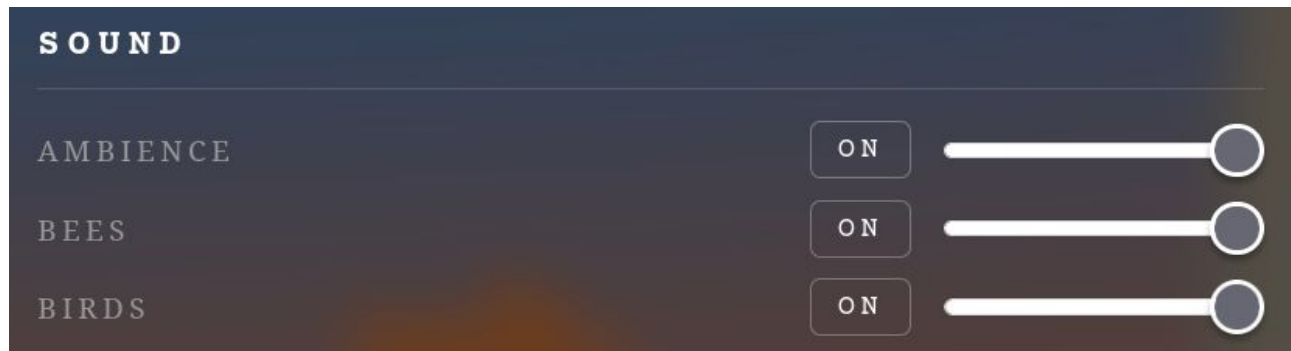


Sounds

→ Ambient sound

→ Bees

→ Birds



User Interaction

User Interaction

→ Camera

- ◆ PerspectiveCamera (FOV 35)
- ◆ Third-person follow camera

→ Keyboard

- ◆ WASD / arrow keys to move the fox
- ◆ Shift to run

→ Mouse

- ◆ Drag to rotate the camera
- ◆ Scroll wheel to zoom in/out (distance is limited)

User Interaction

→ Mobile

- ◆ Virtual joystick to move the fox
- ◆ Button to run
- ◆ One-finger drag to rotate the camera
- ◆ Pinch to zoom

→ Menus

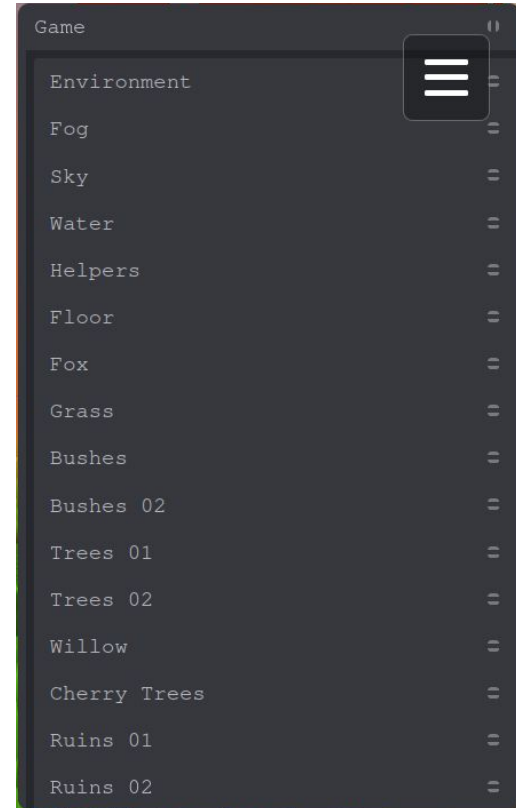
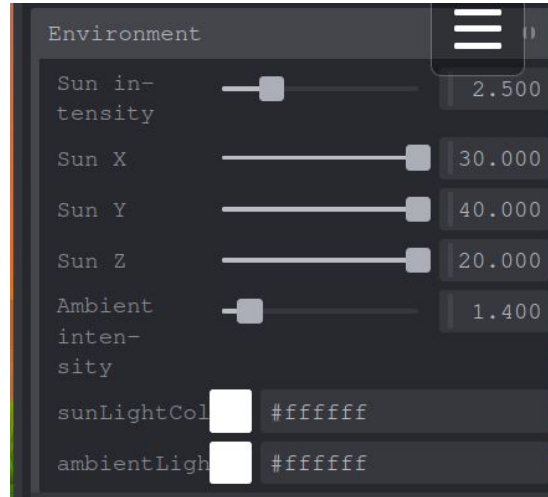
- ◆ Some information about the game
- ◆ Controls
- ◆ Sound customization



User Interaction

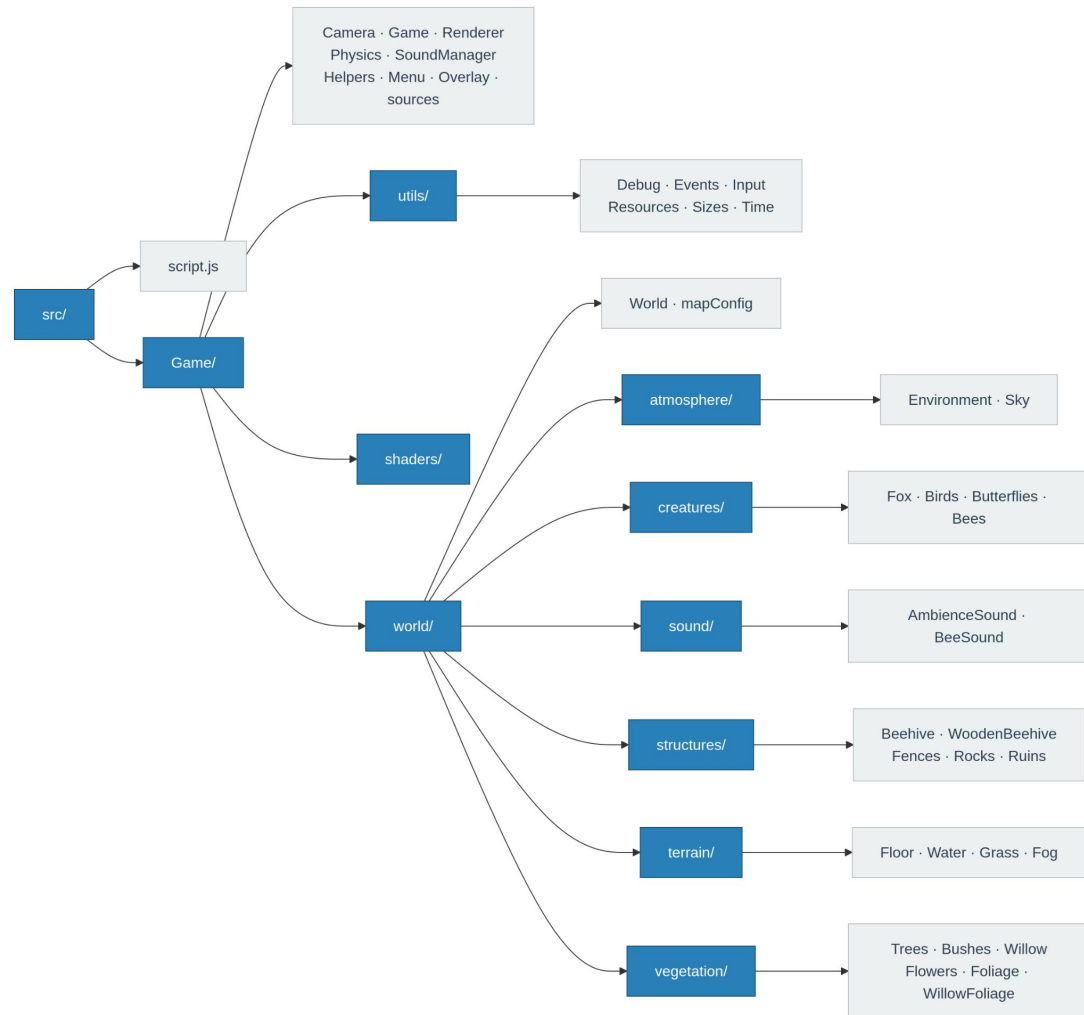
→ Debug Mode

- ◆ Tweak parameters



Development

Code organization



Problems / Difficulties

- Implement all the ideas that came up throughout the semester
- The grass shader was complicated
- Wing flap animation (butterflies)

Use of AI

→ I used Claude to:

- ◆ Discuss implementation approaches
- ◆ Generate repetitive code
 - Add parameters to Tweakpane
 - Random mesh positions
- ◆ Help me on shaders (mainly in the grass shader)
- ◆ Help me on animations (birds, etc.)
- ◆ Loading screen
- ◆ Support for mobile devices and camera movement
- ◆ Debugging issues



Performance

- InstancedMesh is used for trees, bushes, flowers, rocks, etc.
- Custom shaders are better for performance
- Draco compression
- The project runs without restrictions on any Web browser
- The project supports mobile devices

Conclusion

Most important / interesting features

- Vegetation
- Water
- Animals
- Third-person camera

What would you do differently, if starting the project now?

- Think about a different way to position objects around the map
- Improved project planning

References

→ General resources

- ◆ <https://threejs.org/docs/>
- ◆ <https://threejs-journey.com/>
- ◆ <https://www.youtube.com/watch?v=z-XI9tGqH14&t=5052s>
- ◆ <https://google.github.io/draco/>

→ Grass

- ◆ <https://www.youtube.com/watch?v=cesPK0kYkyE&t=360s>

→ Models

- ◆ https://www.youtube.com/watch?v=aM_3b32z8uA
- ◆ <https://www.youtube.com/watch?v=eSL98LLr1kw&t=913s>
- ◆ <https://www.youtube.com/watch?v=DWJ5Wn0Dg7U>